Inequalities — problem 7.13

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Let $\{a_k\}$ $(k = 1, 2, \ldots, n, \ldots)$ be a sequence of pairwise distinct positive integers. Show that for all positive integers n

$$\sum_{k=1}^{n} \frac{a_k}{k^2} \geq \sum_{k=1}^{n} \frac{1}{n} \qquad \text{(IMO 1978)}.$$